



Tab 11 — ADF & Statistical Tests

ADF = Augmented Dickey-Fuller test. Tests whether a time series is **stationary** (mean/variance stable over time) — a fundamental requirement before applying many ML models.

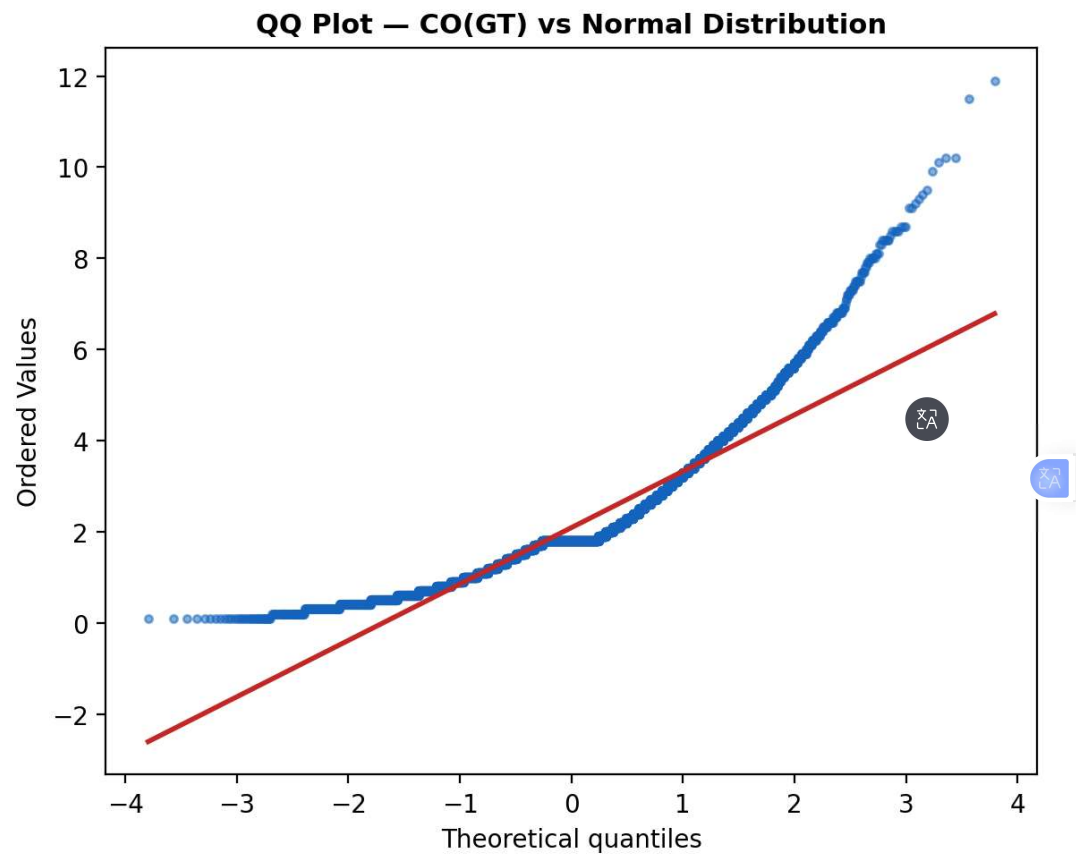
ADF Stationarity Test — All GT Pollutants

	Pollutant	ADF Statistic	p-value	Stationary?	Interpretation
0	CO(GT)	-10.8815	0	Yes	Mean/variance stable
1	C6H6(GT)	-10.2571	0	Yes	Mean/variance stable
2	NOx(GT)	-7.4091	0	Yes	Mean/variance stable
3	NO2(GT)	-8.0878	0	Yes	Mean/variance stable

Normality Test — Shapiro-Wilk (sample n=500)

	Column	W-stat	p-value	Normal?	Skewness	Shape
0	CO(GT)	0.8609	0	No	1.629	Right-skewed
1	C6H6(GT)	0.8793	0	No	1.413	Right-skewed
2	NOx(GT)	0.8175	0	No	2.015	Right-skewed
3	NO2(GT)	0.9559	0	No	0.732	Right-skewed
4	T	0.987	0.0002	No	0.323	Approx. Normal
5	RH	0.9838	0.00002	No	-0.041	Approx. Normal
6	AH	0.9831	0.00002	No	0.265	Approx. Normal

QQ Plot — CO(GT)



✅ All GT pollutants are stationary (ADF $p < 0.05$) — this is good news. Unlike stock prices, pollution levels fluctuate around a stable mean.

✅ None of the pollutants follow a normal distribution — right-skewed due to rare high-pollution events.

⚠️ Non-normality means: use median (not mean) for central tendency, and non-parametric tests when needed.